

Siyuan Gong

📍 Champaign, IL ✉ siyuan3@illinois.edu ☎ +1 312-888-5556 🌐 <https://gooosy.github.io/> 🐙 Gooosy

Education

University of Illinois at Urbana Champaign

Sept 2023 – Present

Bachelor of Science in Computer Engineering

- GPA: 4.0/4.0

Zhejiang University

Sept 2023 – Present

Bachelor of Engineering in Electrical and Computer Engineering

- GPA: 3.90/4.0, Rank: 3/70

Selective Coursework

- CS 225: Data Structures (A+)
- ECE 391: Computer Systems Engineering (A+)
- ECE 220: Computer Systems & Programming (A+)
- ECE 408: Applied Parallel Programming (A+)

Honors and Awards

National Scholarship

Ministry of Education of People's Republic of China, 2024, 2025

Dean's List

University of Illinois at Urbana Champaign, 2024, 2025

Meritorious Winner in Mathematical Contest in Modeling

Consortium for Mathematics and its Applications, 2024

First Prize Scholarship for Academic Excellence

Zhejiang University-University of Illinois Urbana-Champaign Institute, 2024

Gold Medal in iGEM

International Genetically Engineered Machine (iGEM) Foundation, 2024

Research

Graph-Based In-Context Learning Enhancement

Hangzhou, China

Advisors: Yifei Sun, Yang Yang, Zhejiang University

Jun 2024 – Feb 2025

- Conducted literature review, contributed to ideation, and helped refine the model design.
- Designed and implemented experiments: leveraged graph structures and PageRank-based example selection with multi-level encoding (node, path, subgraph), achieving a 2.74% improvement over SOTA baselines.
- Drafted and revised the paper, with emphasis on the experimental methodology and results.

 [Paper](#) 

Competition

International Genetically Engineered Machine Competition

Hangzhou, China

Team: iZJU-China

Apr 2024 – Oct 2024

- Constructed a CNN to detect binary dot matrices in yeast images captured under microscope, facilitating the batch processing of biological images.
- Developed a differential equation model to describe the relationship between yeast genes, enabling prediction of potential product expression levels to assist biological experiments.
- Utilized MATLAB to visualize the results for differential equations and create plots of experimental outcomes.
- Wrote Python scripts to automatically handle batch processing tasks.

 [Project Website](#)   [Gold Medal Certificate](#) 

Mathematical Contest in Modeling

Teammates: Wenjuan Lin, Xinyi Zheng

Hangzhou, China

Feb 2024 – Feb 2024

- Developed the Sea Lamprey Population Model using Fisher's Principle, the Leslie Matrix, and the Lotka-Volterra Model to simulate the dynamics of sea lamprey populations and their impact on ecosystems.
- Utilized MATLAB to stimulate difference equations and visualize the results of the mathematical model through various plots.
- Wrote the paper presenting the final results, addressing how sex ratio variation influences both sea lamprey populations and the surrounding ecosystem.

 [Project Report](#)  [Meritorious Winner Certificate](#) 

Work

Undergraduate Teaching Assistant for ECE 120 Introduction to Computing

Course Instructors: Lin Qiu, Zuozhu Liu, and Ujjal Bhownik

Haining, China

Sep 2024 – Jan 2025

 [Course Wiki](#) 

Undergraduate Teaching Assistant for ECE 408 Applied Parallel Programming

Course Instructors: Volodymyr Kindratenko

Illinois, United States

Sep 2024 – Jan 2025

Course Project

RISC-V Operating System Kernel

Aug 2025 – Dec 2025

- Implemented a RISC-V operating system kernel on QEMU with interrupt/exception handling, threads, and device drivers in C and assembly.
- Developed virtual memory, system calls, ELF loader, and a simple file system supporting user-mode processes.

GPT-2 CUDA Kernel Optimizations

Aug 2025 – Dec 2025

- Optimized CUDA kernels for GPT-2 inference on NVIDIA A40 using Nsight Compute and Nsight Systems profiling.
- Applied warp-level reduction, shared-memory tiling, and split-K parallelism, achieving 20× speedup over baseline.

 [Project Report](#) 

Collaborative Expense Management System

Aug 2025 – Dec 2025

- Built an expense management platform with MySQL on GCP, supporting CRUD, keyword search, multi-currency conversion, and group budgeting.
- Implemented database triggers to maintain group balance consistency and stored procedures with SERIALIZABLE transactions for atomic debt settlement.

 [Project Repo](#) 